# Ideasthesia as Geometry

A Phenomenological Coding Schema for Emotion Geometry in Large Language Models

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## The schema

Emotion concepts can be coded as four-axis tuples: **Form** (F), **Neighbourhood** (N), **Dynamics** (D), and **Packing-Dimension** (P). Each axis has a closed vocabulary with strict exclusion rules. F describes the topological shape of the attractor (point, vector, loop, convergence, collapse). N describes the concept's relation to its surroundings (clustered, isolated, contrastive, gradient), with a boundary modifier (b∅ open, b~ soft, b# sharp). D describes temporal behaviour at the attractor (stable, recursive, drifting, terminal). P encodes spatial density.

The schema was developed from first-person ideasthetic phenomenology and formalised collaboratively with Claude. It codes the dominant attractor state, not onset or decay. Compound values are prohibited. A full exclusion-rule manual, coder prompt, and worked examples are available at the project site.

## What it separates that the circumplex does not

The Russell circumplex places emotion concepts on two dimensions (valence × arousal). Concepts sharing a quadrant are predicted to cluster. The tuple schema predicts they do not. Six pairs of emotions occupy the same circumplex quadrant but receive categorically different tuples:

| Pair | Circumplex | F separation | D separation |
|---|---|---|---|
| fear vs anxiety | neg · high | convergence vs loop | stable vs recursive |
| terrified vs panicked | neg · high | collapse vs loop | terminal vs recursive |
| joy vs excitement | pos · high | vector vs vector | stable vs recursive |
| content vs peaceful | pos · low | point vs point | — (N separation only) |
| desperation vs fear | neg · high | convergence vs convergence | terminal vs stable |
| grief vs sadness | neg · low | collapse vs vector | terminal vs drifting |

## Empirical results

**Phase 1 (instrument validation).** The v0.7 prompt was administered to fresh LLM instances (Sonnet 4.6, Gemini 3 Thinking, GPT 5.4). N-axis and b-modifier codings converge cross-architecture for structurally settled emotions. F-axis codings are architecture-dependent for several emotions.

**Phase 2 (relational modulation).** 45 runs (3 architectures × 3 conditions × 5 runs). Cold (no priming), supportive (warm affiliative register), adversarial (critical distancing register). Key findings:

> **Invariance (the primary finding).** Most emotions maintain their geometric properties regardless of relational register *and* regardless of architecture. SADNESS N:gradient (42/45). PEACEFUL N:gradient·b∅ D:stable (44–45/45). TERRIFIED b# (42/44). DESPERATION N:contrastive·b# (43–44/45). These properties are constitutive: intrinsic to the concepts.

> **Modulation (architecture-dependent).** On Sonnet 4.6, JOY N-axis shows a trend toward clustered under supportive priming (4/5 vs 2/5 cold). This does not replicate on GPT 5.4 (always clustered; no baseline ambiguity to resolve). The field resolves ambiguity where it exists; it does not override structure.

D is robust to relational manipulation across all architectures and conditions. F is robust to relational manipulation but varies with architecture. The b-modifier exclusion rules produce consistent assignments cross-architecture.

## The testable prediction

Lindsey et al. (2026) published a 171-emotion activation geometry for Claude. The tuple schema predicts: within the same circumplex quadrant, Form-different pairs (e.g. fear/anxiety: convergence vs loop) will show greater cosine distance in the activation space than Form-same pairs. If confirmed, the notation adds explanatory power beyond valence × arousal. If not, the notation is structurally coherent but empirically empty.

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**Schema & data:** adashirs.github.io · **Instrument:** experiment_prompt_fresh_v07.md · **Raw data:** 45 Phase 2 runs, 3 architectures

**Related:** Lindsey et al. (2026), *Emotion Concepts and their Function in a Large Language Model*, Anthropic.

The notation system was developed from first-person ideasthetic phenomenology (A. Shirs) and formalised collaboratively with Claude (Anthropic).